Stage F: final report summary

FINAL_V1 - Parkinson's disease detection from speech

Protocol FINAL_V1 Status: **COMPLETED** Generated: 2026-10-20T13:06:52Z

Outcome

- 56 of 56 conditions completed (280 of 280 outer-fold runs); FWL blocked; audit passed.
- Pooled speaker-level AUROC over conditions: 0.867 to 1.000 (median 0.995); AUPRC 0.887 to 1.000.
- Best single condition (descriptive, selection-biased): AC Logistic regression, AUPRC 1.000, AUROC 1.000.

Comparisons

- Hybrids vs constituents: median AUROC difference 0.000 over 76 paired comparisons; interval excludes zero in 5.
- Model variants vs logistic regression: Linear SVM median -0.001; RBF SVM median -0.023; XGBoost median -0.001 (AUROC difference).
- Feature selection vs all 40 acoustic features: median AUROC difference -0.004.

Interpretation limits

- Recording date and class are confounded; acoustic features alone reach median AUROC 0.996. Scores are internal and do not show disease detection.
- Errors concentrate in a few speakers: PD_005 (54 of 56 conditions); HC_001 (14 of 56 conditions); HC_020 (11 of 56 conditions); HC_006 (5 of 56 conditions).

Table 1: Median pooled AUROC by representation (over models)

Representation	Median	Min	Max
AC	0.999	0.989	1.000
H3	0.999	0.973	1.000
H8	0.997	0.975	0.998
AC/FORWARD	0.996	0.996	0.996
XLSR	0.996	0.976	1.000
WLM	0.995	0.873	0.998
H6	0.994	0.975	1.000
H7	0.994	0.971	0.996
H5	0.994	0.975	0.998
H1	0.993	0.867	0.998
H4	0.991	0.975	0.995
AC/BACKWARD	0.990	0.985	0.995
H2	0.989	0.962	0.998
W2V	0.985	0.975	0.996